

LONGZHOU, China

WMO: 594170

Lat: 22.3398N

Lon: 106.8536E

Elev: 423

StdP: 14.47

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	42.7	45.7	31.2	25.8	56.9	34.9	30.2	55.1	11.0	64.2	9.7	62.0	2.2	110	0.282

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.2	96.8	79.6	95.0	79.4	93.3	79.0	82.2	91.6	81.5	90.3	80.8	89.2	4.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.0	158.3	85.8	79.3	154.5	85.2	78.8	151.8	84.8	46.6	92.0	45.8	90.7	45.1	89.6	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.5	7.8	7.1	38.4	100.9	2.7	1.9	36.4	102.3	34.8	103.4	33.3	104.5	31.3	105.9		
(5)			37.6	83.7	2.5	1.1	35.8	84.4	34.4	85.1	33.0	85.7	31.2	86.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	73.9	58.5	62.3	67.5	76.0	81.4	84.0	84.2	83.7	81.5	76.7	69.2	61.0	(6)
(7)		DBStd	10.85	7.22	8.63	7.49	6.66	4.82	3.24	3.05	2.96	3.57	4.55	5.51	6.22	(7)
(8)		HDD50	13	8	3	0	0	0	0	0	0	0	0	0	2	(8)
(9)		HDD65	623	230	146	60	6	0	0	0	0	0	1	22	158	(9)
(10)		CDD50	8735	271	347	543	781	975	1019	1062	1045	946	827	575	344	(10)
(11)		CDD65	3870	28	70	138	337	510	569	597	580	496	363	147	35	(11)
(12)		CDH74	37199	104	394	736	2702	5266	6492	6785	6212	4817	2715	818	158	(12)
(13)		CDH80	14731	12	107	210	933	2221	2777	2898	2563	1890	912	187	21	(13)
(14)	Wind	WSAvg	2.8	3.1	3.3	3.1	3.1	2.8	2.6	2.7	2.5	2.4	2.4	2.5	2.8	(14)
(15)	Precipitation	PrecAvg	51.7	1.1	1.2	1.9	3.9	7.0	8.7	8.3	8.7	5.0	3.5	1.6	0.7	(15)
(16)		PrecMax	69.1	3.5	4.8	6.8	8.3	15.9	22.4	17.6	17.4	10.0	8.3	5.5	3.6	(16)
(17)		PrecMin	37.4	0.0	0.0	0.3	0.2	2.2	1.6	2.6	1.9	2.5	0.1	0.0	0.0	(17)
(18)		PrecStd	7.6	0.9	1.1	1.6	2.0	3.2	4.2	3.4	4.1	1.9	2.0	1.5	0.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	81.7	88.6	91.2	96.9	99.4	98.3	98.3	97.2	95.7	92.1	87.9	82.8	(19)
(20)			MCWB	68.2	69.7	73.2	76.2	79.3	80.8	80.8	79.7	78.2	75.0	73.0	70.8	(20)
(21)		2%	DB	76.8	83.2	85.5	91.8	95.7	95.7	95.8	95.2	93.4	89.7	84.7	78.4	(21)
(22)			MCWB	66.5	69.8	73.0	76.9	79.0	80.4	80.2	79.7	77.7	74.7	71.7	67.9	(22)
(23)		5%	DB	73.0	78.9	81.2	88.5	92.8	93.4	93.8	93.2	91.4	87.6	82.0	74.6	(23)
(24)			MCWB	64.5	68.4	71.7	76.1	78.1	80.0	79.9	79.7	77.5	73.6	69.4	64.6	(24)
(25)		10%	DB	69.5	74.8	77.6	85.3	90.0	91.2	91.6	91.0	89.3	85.4	78.9	71.2	(25)
(26)			MCWB	63.0	67.0	70.2	74.9	77.3	79.6	79.8	79.5	76.7	72.9	68.5	62.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.9	72.5	75.6	79.6	82.2	82.9	83.1	82.8	81.5	78.7	76.1	72.7	(27)
(28)			MCD B	77.2	82.7	85.4	89.8	92.7	92.7	93.0	92.0	89.5	87.6	82.9	80.3	(28)
(29)		2%	WB	67.7	70.8	73.9	78.1	80.8	81.9	82.0	81.8	80.3	77.3	74.4	69.9	(29)
(30)			MCD B	75.0	79.5	82.0	87.8	90.8	91.0	91.2	90.7	88.1	84.8	80.7	75.8	(30)
(31)		5%	WB	66.1	69.3	72.7	76.9	79.8	81.2	81.2	81.0	79.5	76.3	72.9	67.3	(31)
(32)			MCD B	71.7	76.9	79.7	85.6	88.9	89.7	89.8	89.2	86.9	83.0	78.6	72.5	(32)
(33)		10%	WB	64.2	67.8	71.3	75.9	78.9	80.5	80.5	80.2	78.7	75.3	71.3	64.5	(33)
(34)			MCD B	68.8	74.3	77.1	83.7	87.0	88.4	88.4	87.7	85.6	81.2	76.7	69.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.6	11.9	10.8	12.4	13.7	13.1	13.2	13.6	14.1	14.7	15.1	14.4	(35)
(36)			MCDBR	19.1	19.3	17.4	17.4	18.4	16.9	17.1	17.4	17.3	18.0	19.6	19.2	(36)
(37)			MCWBR	8.7	7.2	6.3	5.4	4.7	4.4	4.5	4.7	4.3	5.0	7.3	8.7	(37)
(38)		5% WB	MCDBR	15.0	17.2	14.6	15.3	15.4	14.3	14.7	15.3	14.4	14.6	14.5	15.8	(38)
(39)			MCWBR	7.4	7.2	6.3	5.4	5.1	4.9	4.9	5.2	4.4	5.0	6.5	8.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.597	0.621	0.792	0.746	0.561	0.542	0.513	0.577	0.617	0.711	0.595	0.620	(40)	
(41)		taud	1.718	1.706	1.432	1.556	2.008	2.076	2.150	1.948	1.818	1.586	1.776	1.689	(41)	
(42)		Ebn at Noon	210	217	188	202	241	244	251	236	224	193	210	197	(42)	
(43)		Edh at Noon	68	73	100	90	57	53	49	60	67	82	64	68	(43)	
(44)	All-Sky Solar Radiation	RadAvg	656	800	878	1144	1374	1424	1493	1479	1403	1164	981	776	(44)	
(45)		RadStd	165	153	118	134	145	158	114	97	95	93	145	136	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	+1.03	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	N/A	N/A	N/A	+0.94	N/A	N/A	N/A	N/A	N/A	+131

Nomenclature: See separate page